

Skills Summary

Evaluating Functions from Rules, Tables, and Graphs

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluate a function from its rule

Rule: $f(\text{input}) = \text{output}$. To find $f(a)$, replace every x in the rule with a — in parentheses — then simplify.

- $f(3)$ is *not* f times 3. The parentheses hold the input.
- Keep the input in brackets when it is negative: $(-4)^2 = 16$, but $-4^2 = -16$.
- Order of operations still applies after substituting.

Example: $f(x) = 4x - 7$. Find $f(3)$.

Step 1. The rule is given, so evaluating means substituting the input for x and simplifying — no table or graph is needed.

Step 2. $f(3) = 4(3) - 7 = 12 - 7. \Rightarrow 5$

Try it: $f(x) = x^2 + 2x$. Find $f(-4)$.

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Skill 2 — Read a value from a table or a graph

Rule: A table row and a graph point say the same thing as a rule: they pair one input with one output.

- **Table:** find the input in the input row, read the output directly below.
- **Graph:** find the input on the x -axis, go up (or down) to the curve, then across to the y -axis. That height is the output.
- If a question asks “for which input”, the answer is an x -value, not a height.

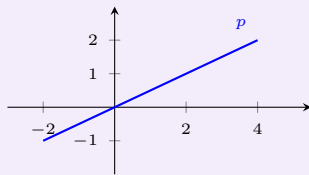
Example: Find $q(2)$ from this table.

x	0	1	2	3
$q(x)$	2	5	8	11

Step 1. The table already lists the outputs, so reading replaces computing — locate the input column instead of substituting.

Step 2. The column headed 2 has 8 directly beneath it. $\Rightarrow 8$

Try it: Read $p(4)$ from the graph of p .



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Skill 3 — Compare two functions at the same input

Rule: To compare f and g , evaluate **both at the same input**, then compare the two outputs.

- On a graph: at that input, the higher curve has the greater output.
- Where the graphs cross, the outputs are equal — that input solves $f(x) = g(x)$.
- Comparing at *different* inputs answers a different question.

Example: $f(x) = 4x - 7$ and $g(x) = 2x + 1$. Compare $f(3)$ and $g(3)$.

Step 1. Same input, two rules — evaluate each separately, then compare the outputs.

Step 2. $f(3) = 4(3) - 7 = 5$ and $g(3) = 2(3) + 1 = 7$; 5 is less than 7. $\Rightarrow f(3) < g(3)$

Try it: $f(x) = 5x - 3$ and $g(x) = x^2 + 2$. Compare $f(2)$ and $g(2)$.

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